Getting Started

SWAT+ Documentation

docs.swat.tamu.edu

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Getting Started

This section gets you from a fresh machine to a running SWAT+ simulation.

If you have never run the model before, follow the pages in order:

1. Install. Get the SWAT+ executable on your machine.
2. Quick start: Ames_sub1. Run the small reference dataset shipped with the source repository.
3. Project folder anatomy. Understand what is in a SWAT+ project directory and what each group of files does.
4. Your first watershed. Build a project for a new area using QSWAT+ and run it.

If you only want to read about the model first, skip ahead to Model theory.

SWAT+ is a watershed model. It simulates water, sediment, nutrient, pesticide, and crop yield response to climate and management at scales from a single field up to a large river basin.

This section covers what the model is, how it differs from the original SWAT, how to cite it, and how releases are numbered.

Pages

- What is SWAT+
- SWAT vs SWAT+
- Citing and license
- Release notes

The Soil and Water Assessment Tool Plus (SWAT+) is an open-source watershed model developed jointly by the USDA Agricultural Research Service (USDA-ARS) and Texas A&M AgriLife Research, with contributions from Colorado State University, the USDA-ARS Hydrology and Remote Sensing Laboratory and other institutions.

SWAT+ simulates the quantity and quality of surface water and groundwater, from small watersheds to large river basins. It is used to evaluate the effects of land use, land management, soils and climate change on water resources.

Background

Over more than two decades the original Soil and Water Assessment Tool (SWAT) became one of the most widely used watershed models in the world. Its many applications showed limits in how a watershed could be represented, and the additions made over the years left a code base that was hard to maintain and extend. SWAT+ is a restructured version of SWAT. Most of the process equations are the same, but the code is organized around objects and the input files are relational, which makes the model easier to maintain and lets new science be added as separate modules. SWAT+ also allows a more flexible spatial representation: landscape units, HRUs, aquifers, channels and reservoirs can be connected in any arrangement through connection files. The main differences are listed on SWAT vs SWAT+.

Processes simulated

For each HRU, channel, aquifer, reservoir and wetland, SWAT+ keeps a daily mass balance of:

- water: precipitation, snow, evapotranspiration, surface runoff, lateral flow, percolation, groundwater flow and channel routing
- sediment: detachment, transport and deposition on the landscape and in channels
- nutrients: nitrogen and phosphorus cycling in the soil, plant uptake, transport with runoff and subsurface flow, and in-stream transformations
- carbon: soil organic carbon and residue pools
- pesticides and pathogens: application, decay and transport
- salts and other constituents
- plant growth: biomass, leaf area, yield, and water, nutrient and temperature stress

Management (planting, tillage, fertilizer and manure, irrigation, harvest, grazing) can follow fixed schedules or rules in decision tables. Water can be allocated between sources and users, and structural practices such as tile drains, filter strips and terraces can be represented.

Output can be written daily, monthly, yearly and as average annual values. The Model Theory section describes each process.

Applications

- effects of management practices on water yield and water quality
- soil erosion and non-point source pollution
- crop yield response to climate and management
- watershed planning and total maximum daily load (TMDL) studies
- climate change impacts on water resources
- irrigation and water allocation

Running the model

SWAT+ is a command-line program. It reads a folder of plain-text input files that describe the watershed, climate, soils, land use, management and simulation period, and writes plain-text (and optionally CSV) output files to the same folder.

Most users do not write the input files by hand. They use the supporting tools:

- **QSWAT+**, a QGIS plugin, delineates the watershed from a DEM, creates HRUs from soil, land use and slope maps, and writes the project database.
- **SWAT+ Editor** edits the inputs of a project, writes the input files and runs the model.
- **SWAT+ Toolbox** runs sensitivity analysis, calibration and other tasks on a finished project.

All three can be installed from install.swat.tamu.edu. See Install.

Source and license

The source code is at github.com/swat-model/swatplus and is released under the GNU Lesser General Public License 2.1. See Citing and license.

Sources

- `README.md` and `CITATION.cff` in the source repository at 62.0.1.
- Input file groups: `src/input_file_module.f90`.

SWAT+ is the restructured successor to SWAT (Soil and Water Assessment Tool). Most of the process equations are the same. The code structure, the input files and the way parts of a watershed are connected are different. The SWAT column below refers to SWAT2012, the last release of the original model (`swat-model/swat2012`).

Spatial structure

	SWAT2012	SWAT+
Units	subbasins, each with HRUs and one main channel	landscape units, routing units, HRUs, channels, aquifers, reservoirs, wetlands, point sources, each a separate object
Connections	fixed: HRUs drain to their subbasin's channel, subbasins drain downstream in numbered order	defined in connection files (`*.con`): any object can send any part of its flow (surface runoff, lateral flow, tile flow, recharge) to any other object
Floodplain and upland	not separated	landscape units can separate upland and floodplain areas

The connection files make it possible, for example, to route upland runoff across a floodplain before it reaches the channel, to send tile flow from one HRU to a wetland, or to divide a reservoir's outflow between two channels.

Input files

SWAT2012 used one set of files per subbasin and HRU (`.sub`, `.hru`, `.mgt`, `.gw`, `.rte`, `.sol`, …), each repeating the full parameter list.

SWAT+ uses relational input files grouped by purpose. A parameter set is written once, given a name, and referenced by name from every object that uses it. For example, one row of `hydrology.hyd` can serve hundreds of HRUs. The master file `file.cio` lists every input file by group:

- simulation: `time.sim`, `print.prt`, `object.cnt`
- basin: `codes.bsn`, `parameters.bsn`, `carbon.bsn`
- climate: `weather-sta.cli`, `weather-wgn.cli`, `pcp.cli`, `tmp.cli` and the station files
- connections: `hru.con`, `rout_unit.con`, `aquifer.con`, `chandeg.con`, `reservoir.con`, `recall.con`, …
- HRUs, land use and management: `hru-data.hru`, `landuse.lum`, `management.sch`, `cntable.lum`, decision tables (`*.dtl`)
- hydrology: `hydrology.hyd`, `topography.hyd`, `field.fld`
- soils and plants: `soils.sol`, `nutrients.sol`, `plants.plt`, `plant.ini`
- databases: fertilizer, tillage, pesticide, urban and other parameter tables

The Input Reference describes each file.

Management

SWAT2012 scheduled operations in the `.mgt` file of each HRU, by date or by fraction of heat units.

SWAT+ management schedules (`management.sch`) are shared between HRUs, and can combine dated operations with decision tables (`*.dtl`). A decision table checks conditions every day (soil water, plant growth stage, date, crop rotation year, reservoir level, and so on) and applies an action when they are met. Decision tables also drive reservoir releases, flow control and land use change. See Decision tables.

Groundwater

SWAT2012 had a shallow and a deep aquifer for each HRU, with the shallow aquifer feeding the main channel of its subbasin.

SWAT+ aquifers are separate objects, usually one per landscape unit or group of HRUs, connected to channels through `aquifer.con`. SWAT+ also includes gwflow, a gridded groundwater module that simulates water table heads and exchanges with channels, soils, tile drains, reservoirs and wells cell by cell. See Aquifers.

Additional processes

SWAT+ adds or extends: a carbon module with soil organic matter pools, water allocation between sources and users, salts and other constituents, rice paddies, and a sediment budget for channels. Each is described in the Model

Theory section.

Outputs

SWAT2012 wrote a fixed set of files (`output.hru`, `output.sub`, `output.rch`, `output.std`).

SWAT+ writes one file per object type, process and time step, switched on and off in `print.prt`. Examples from the `Ames_sub1` reference dataset:

```
basin_wb_aa.txt        basin water balance, average annual
hru_wb_aa.txt          HRU water balance, average annual
channel_sd_day.txt     channel flow and loads, daily
basin_aqu_aa.txt       aquifer, average annual
```

The suffix gives the time step: `_day`, `_mon`, `_yr` or `_aa`. Each row of `print.prt` switches the four time steps on or off. See the Output Reference.

Code

	SWAT2012	SWAT+
Language	Fortran, fixed form	Fortran, free form
Organization	shared common blocks	modules grouped by process and object
Source	released as archives	public on GitHub, LGPL-2.1
Build	hand-written makefiles	CMake

Migration

There is no automatic converter from a SWAT2012 project to SWAT+. Rebuild the project in QSWAT+ from the same DEM, soil, land use and climate data, then transfer calibrated parameter values by hand, checking each against the SWAT+ units and meaning.

Sources

- Connection and input file names: `type input_con` and the other `input_*` types in `src/input_file_module.f90`.
- Aquifer objects: `src/aquifer_module.f90`, read by `aqu_read.f90`. Gridded groundwater: `gwflow_*.f90`. Spatial aquifer connections (`aquifer2d.con`): `aqu2d_read.f90`, `aqu2d_init.f90`.
- Build: `CMakeLists.txt` compiles `src/*.f90` free form (`-free` for Intel, `-ffree-line-length-none` for GNU) into one executable.
- SWAT2012: `swat-model/swat2012`.

License

SWAT+ is released under the GNU Lesser General Public License, version 2.1 (LGPL-2.1). The full text is in the `LICENSE` file of the source repository.

Citing the software

Cite the model release used in the study, with its version and DOI. The author list below is the one in the repository's citation file at 62.0.1:

> Arnold, J. G., Čerkasova, N., White, M. J., Bailey, R., Thorp, K., Jeong, J., Zhang, X., Ugraskan, T., van Griensven, A., Rathjens, H., Raj, C., Cai, X., Geter, W. F., David, O., Carlson, J. R., and Chawanda, C. J. Soil and Water Assessment Tool Plus (SWAT+), version 62.0.1. https://doi.org/10.5281/zenodo.14983533

Releases are archived on Zenodo:

DOI	Points to
10.5281/zenodo.14983533	all versions; resolves to the newest archived release
10.5281/zenodo.22919722	release 62.0.0
10.5281/zenodo.14983534	release 61.0.2

When a release has a version DOI, cite that DOI so the exact code can be found again.

GitHub shows a "Cite this repository" button on the repository page, built from the same citation file, with APA and BibTeX forms.

Citing the documentation

When a statement comes from this site, cite the page and the documentation version shown in the version selector at the top of the page, for example "SWAT+ Documentation, revision 62.0.1, Soft calibration, https://docs.swat.tamu.edu/62.0.1/calibration/soft-calibration/".

Older versions

Revisions before the move to GitHub are in the modular_swatplus repository on Bitbucket.

Sources

- `CITATION.cff`, `LICENSE` and `README.md` at 62.0.1.
- Zenodo records 14983533 (concept), 22919722 and 14983534, checked 20 October 2026.

SWAT+ releases are listed on the GitHub releases page, with prebuilt executables and the merged pull requests of each release. Version numbers are explained in Tagging and releases.

This page lists the changes in each release that affect model users: input files, outputs and results. Each version of this documentation describes one release; use the version selector at the top of the page for earlier ones.

62.0.1

Released 19 October 2026.

Input files

- `file.cio`, `water_rights` line: now names two files, the places of use (`place_of_use.wro`) and the points of diversion (`point_of_diver.wro`), in place of the three water rights files. The water allocation input files were reorganized around places of use, points of diversion and return; see Water allocation. A `file.cio` written for 62.0.0 or earlier must be updated; see Older file.cio files.
- `parameters.bsn`: two columns that were unused are now read as the pesticide groundwater mixing factor and a switch for temperature-dependent pesticide decay. SWAT+ Editor still labels them `lin_sed` and `exp_sed`; a value of 1 in the second turns temperature-dependent decay on. See parameters.bsn.
- `pesticide.pes`: three new columns for plant uptake. Files in the older layout are still read, with no plant uptake. See pesticide.pes.
- Recall: the recall file named in `file.cio` is read (instead of a fixed name), and empty recall files are no longer created. See recall.
- `exco.exc` and `exco_om.exc`: the original layout is restored.
- Manure: a manure database (`manure_db.frt`, `manure_om.frt`) and manure allocation inputs were added. See manure allocation.
- Transplanting: transplant data are read from `transplant.ops`; `transplant.plt` is no longer read.

Outputs

- Water allocation outputs were replaced by place-of-use based files.
- New sediment budget outputs for channels, stream orders and the basin, with corrected average annual values.
- Rice paddy output, and paddy water included in the ET reports.
- gwflow outputs now follow their `print.prt` and gwflow switches, and the average annual storage change is computed correctly.
- The legacy carbon output no longer writes duplicate rows for each HRU.

See the Output Reference for the files.

Model changes

- Rice paddies: seepage, weir spill volume and the impoundment switch day were revised, and soil evaporation is skipped while paddy water remains.
- Canopy interception is applied for all runoff methods.
- Pesticides: temperature-dependent decay, plant uptake and mass balance corrections.
- Water allocation and irrigation were reworked; the irrigation days of an unlimited source are reset correctly.

Calibration

- `aquifer_delay` (gwflow) is changed in days.
- New object types `gwf_riv` and `gwf_sgl` for gwflow channel and single-value parameters.
- `nperco_lchtile` changes the right parameter.

See Parameter change files.

Fixes

- A crash in tillage mixing for plant communities with more than 12 plants.
- An endless loop when all operations of a management schedule share one date.
- Root depth of perennials stuck at 25.4 mm at a cold start.
- Release executables built with gfortran no longer stop with a floating-point trap on macOS, and the macOS executable no longer needs Homebrew's gcc libraries.

Documentation for 62.0.1

This version of the documentation was rewritten against the 62.0.1 source. Pages now describe what the model does, including inputs that are read but have no effect and behaviour that differs from the input file labels, so that results can be interpreted correctly. Examples:

- Measured PET is replaced by the Hargreaves estimate within the same day, so `pet` 3 in `codes.bsn` gives Hargreaves results (codes.bsn).
- Name lookups between files fall back to the first entry without a message when a name does not match; check `diagnostics.out` and the names themselves (weather-sta.cli, hru-data.hru).
- Several basin parameters cannot be changed through `calibration.cal` (Parameter change files).
- Manure allocation, street sweeping and the pathogen simulation have no effect, and water allocation behaves as its inputs describe only for a single place of use (Water allocation).

Each page states these points where they apply.

Sources

- SWAT+ 62.0.1 release and `git log 62.0.0..62.0.1` in the source repository.
- Input changes: `type input_water_allocation` in `src/input_file_module.f90` and `read (107,*) name, in_wallo` in `src/readcio_read.f90`; `pestgwfact` and `temp_decay` replacing `spcon` and `spexp` in `src/basin_module.f90`; `src/pesticide_data_module.f90`; `src/recall_read.f90` and `src/recalldb_read.f90`; `src/exco_db_read.f90` and `src/exco_read_om.f90`; `src/plant_transplant_read.f90`.

- Outputs: `src/wallo_pou_output.f90`, `src/basin_sedbud_output.f90`, `src/ch_sedbud_output.f90`, `src/chord_sedbud_output.f90`, `src/gwflow_output.f90`.
- Build: `CMakeLists.txt` (gfortran flags and macOS linking).

SWAT+ has two parts: the model itself, a command-line program, and the tools used to build and edit projects (QSWAT+, SWAT+ Editor and SWAT+ Toolbox). Most users install the tools and let SWAT+ Editor run the model. The model can also be downloaded or built without the tools.

SWAT+ tools

install.swat.tamu.edu has one-line installers for Windows, Linux and macOS. Each installer can be run again at any time to update.

Install all three tools at once:

```
# Linux and macOS
curl -fsSL https://install.swat.tamu.edu/all.sh | bash

# Windows (PowerShell)
irm https://install.swat.tamu.edu/all.ps1 | iex
```

Or one at a time:

Tool	Purpose	Linux and macOS	Windows (PowerShell)
QSWAT+	QGIS plugin that delineates the watershed, creates HRUs and writes the project database. Also installs the SWAT+ support data and TauDEM. Needs QGIS.	`curl -fsSL https://install.swat.tamu.edu/qsp.sh | bash`	`irm https://install.swat.tamu.edu/qsp.ps1 | iex`
SWAT+ Editor	Edits project inputs, writes the input files and runs the model.	`curl -fsSL https://install.swat.tamu.edu/spe.sh | bash`	`irm https://install.swat.tamu.edu/spe.ps1 | iex`
SWAT+ Toolbox	Sensitivity analysis, calibration and other tasks on a finished project.	`curl -fsSL https://install.swat.tamu.edu/spt.sh | bash`	`irm https://install.swat.tamu.edu/spt.ps1 | iex`

On Linux and macOS the tools go to `~/.local/share/SWATPlus/`, and on Windows to `%USERPROFILE%\SWATPlus\`.

Model executable

Every release on the GitHub releases page has prebuilt executables. For 62.0.1:

File	System
`swatplus-62.0.1-gnu-win_amd64-Rel.zip`	Windows, 64-bit (gfortran build)
`swatplus-62.0.1-ifo-win_amd64-Rel.zip`	Windows, 64-bit (Intel Fortran build)
`swatplus-62.0.1-gnu-lin_x86_64-Rel.zip`	Linux, x86-64 (gfortran build)
`swatplus-62.0.1-ifx-lin_x86_64-Rel.zip`	Linux, x86-64 (Intel Fortran build)
`swatplus-62.0.1-gnu-lin_aarch64-Rel.zip`	Linux, ARM64
`swatplus-62.0.1-gnu-mac_arm64-Rel.zip`	macOS, Apple silicon
`swatplus-62.0.1-ifo-mac_x86_64-Rel.zip`	macOS, Intel

Each zip contains a single executable with the same name (plus `.exe` on Windows). The name tells you the version, the compiler (`gnu` for gfortran, `ifo` for Intel Fortran Classic, `ifx` for Intel Fortran), the system and architecture,

and `Rel` for an optimized release build. The gfortran and Intel builds give results that can differ slightly in the last digits.

Unzip the file and put the executable next to your project folder or in a folder on your `PATH`. On Linux and macOS, make it executable:

```
chmod +x swatplus-62.0.1-gnu-lin_x86_64-Rel
```

On macOS the first start of a downloaded program may be blocked; allow it under System Settings, Privacy and Security, or remove the quarantine flag with `xattr -d com.apple.quarantine <file>`.

To check the download, run it in an empty folder:

```
mkdir empty && cd empty
../swatplus-62.0.1-gnu-lin_x86_64-Rel
```

It prints a banner with the revision number and stops because there is no project:

```
                  SWAT+
             Revision 62.0.1
      Soil & Water Assessment Tool
GNU (13.3.0), 2026-10-19 22:51:37, Linux
    Program reading . . . executing

 Cannot find object.cnt input file
```

It also writes a few files (`diagnostics.out`, `simulation.out`, `codes.bsn` and others) into the folder. Run the model from inside a project folder; it reads its inputs from the current folder and writes the outputs there.

Building from source

Build from source to change the model or to use a development version. You need `git`, CMake 3.22 or newer, a Fortran compiler (gfortran, Intel `ifx` or `ifort`) and `make` or `ninja`.

```
git clone https://github.com/swat-model/swatplus.git
cd swatplus
git checkout 62.0.1
cmake -B build -D CMAKE_BUILD_TYPE=Release
cmake --build build -j 8
```

The executable is written to `build/` with a name of the form `swatplus-<version>-<compiler>-<system>_<arch>-Rel` (`-Dbg` for a `Debug` build, no suffix when no build type is given). Builds from a commit between tags have a version such as `62.0.1-3-g1a2b3c4`. The Developer Guide has instructions for Visual Studio, VS Code and Codespaces.

Next

Run the reference dataset to check the executable from start to finish: Quick start: Ames_sub1.

Sources

- Release files: SWAT+ 62.0.1 on GitHub, checked 20 October 2026; the banner above is from the `gnu-lin_x86_64` executable.
- Installers: install.swat.tamu.edu, checked 20 October 2026.
- Executable naming: `set(SWATPLUS_EXE "swatplus-${SWAT_VERSION}-${FFC}-${AR}${TY}")` in `CMakeLists.txt`, with `SWAT_VERSION` from `git describe --tags`.
- Build instructions: `doc/Building.md`.

Run a complete SWAT+ simulation with the small reference dataset that comes with the source code.

The dataset

`Ames_sub1` is a test dataset of 12 HRUs near Ames, Iowa, with no channels, aquifers or reservoirs. It simulates 1 January 1975 to 31 December 2020 on a daily time step. It is used to test the model and has not been calibrated. The folder is in the SWAT+ source repository at `refdata/Ames_sub1`.

Before running

You need:

- the SWAT+ executable (see Install);
- a copy of the `Ames_sub1` folder, either from a clone of the source repository or downloaded from the link above;
- about 10 GB of free disk space. With the shipped `print.prt` the run writes daily carbon output for every HRU, about 9 GB in 62.0.1. To keep the output small, change the `daily` column of the `hru_cb` and `hru_cb_vars` rows in `print.prt` from `l` to `n` before running.

The shipped folder already contains output files from earlier model versions (for example `basin_wb_aa.txt`, written in 2024). A run with the shipped `print.prt` does not rewrite them, so check the date in the first line of an output file, or delete the old outputs (`*_aa.txt`, `*_yr.txt`, `*_day.txt`, `*.out`) from your copy before running.

Running

From a terminal, change into your copy of the folder and run the executable from there:

```
# Linux and macOS
cd path/to/Ames_sub1
path/to/swatplus-62.0.1-gnu-lin_x86_64-Rel
```

```
:: Windows
cd path\to\Ames_sub1
path\to\swatplus-62.0.1-gnu-win_amd64-Rel.exe
```

SWAT+ reads its inputs from the current folder, so always start it from inside the project folder. The run takes a few minutes; on a recent Linux workstation it took 4 minutes 20 seconds, most of it writing the daily carbon output.

Output

While it runs, SWAT+ prints the simulation date to the screen:

```
  Original Simulation            1   14   1994 Yr    20 of    46 Time   12:37:36
```

At the end, the screen and `simulation.out` show:

```
 Execution successfully completed
```

and the same line is written to `success.fin`.

With the shipped `print.prt`, a 62.0.1 run writes:

Files	Content
`hru_wb_day.txt`, `hru_wb_aa.txt`	HRU water balance, daily and average annual
`hru_nb_day.txt`, `hru_nb_aa.txt`, `hru_ncycle_day.txt`, `hru_ncycle_aa.txt`	HRU nutrient balance and nitrogen cycle
`hru_ls_aa.txt`	HRU sediment and nutrient losses, average annual
`hru_pw_day.txt`, `hru_pw_aa.txt`	HRU plant growth and stress
`crop_yld_yr.txt`, `crop_yld_aa.txt`, `basin_crop_yld_yr.txt`, `basin_crop_yld_aa.txt`	crop yields

Files	Content
`hru_carbvars.txt`, `hru_cbn_lyr.txt`, `hru_seq_lyr.txt`, `hru_cflux_stat.txt`, `hru_cpool_stat.txt`, `hru_n_p_pool_stat.txt`, `hru_org_*_vars.txt`, `hru_plc_stat.txt`, `basin_carbon_all.txt`, `hru_begsim_soil_prop.txt`, `hru_endsim_soil_prop.txt`	carbon and soil organic matter
`mgt_out.txt`, `lu_change_out.txt`	management operations performed, land use changes
`simulation.out`, `diagnostics.out`, `checker.out`, `erosion.out`, `files_out.out`, `co2.out`	run log, warnings, input summaries

A few other files appear as well: `paddy_test.csv` (a rice paddy test output written on every run), `wtps_wuses.wal` and an empty file named `null`.

Open `hru_wb_aa.txt` to see the average annual water balance of each HRU. For the first HRU in this run:

precip	surq_gen	latq	perc	et
952.6 mm	34.7 mm	0.1 mm	0.0 mm	932.5 mm

The columns are described in Water balance outputs.

Warnings in diagnostics.out

`diagnostics.out` lists anything the model could not match while reading the inputs. For `Ames_sub1` at 62.0.1 it contains:

```
agrr_lum      AGRR_rot      not found in management.sch
soyb_lum      SOYB_rot      not found in management.sch
oats_lum      OATS_rot      not found in management.sch
wwht_lum      WWHT_rot      not found in management.sch
ames.tem                    file not found (tgage)
```

The first four are land uses in `landuse.lum` that name management schedules missing from `management.sch`; no HRU uses those land uses. The last line means the station's temperature file, `ames.tem` in `weather-sta.cli`, is not in the list in `tmp.cli` (which names `ames.tmp`), so the model generates daily temperatures from the weather generator statistics in `weather-wgn.cli`. The Ames_sub1 walkthrough explains both. Check this file after every first run of a new project.

Troubleshooting

- `Cannot find object.cnt input file`: SWAT+ was started outside the project folder. Change into the folder first.
- `Permission denied` on Linux or macOS: run `chmod +x` on the executable.
- A command or the model stops with no message: check free disk space.
- The run stops early: read `simulation.out`, `diagnostics.out` and `checker.out` for the last message before the stop.

See FAQ and troubleshooting for other errors.

Next

- Project folder anatomy explains what each file in the folder does.
- The Ames_sub1 walkthrough goes through the dataset file by file.
- Your first watershed builds a project for a new area.

Sources

- Dataset: refdata/Ames_sub1 at 62.0.1 (`object.cnt`: 12 HRUs and no other objects; `time.sim`: 1975 to 2020, daily; `print.prt`: HRU rows only, with `hru_cb` and `hru_cb_vars` daily set to `1`).
- Run: `swatplus-62.0.1-gnu-lin_x86_64-Rel` from the 62.0.1 release, 20 October 2026. Output size and file list are from that run.
- Completion message and `success.fin`: src/main.f90.in. `paddy_test.csv` is opened unconditionally in `main.f90.in` as a temporary paddy output.

A SWAT+ project is one folder of plain-text files, usually called `TxtInOut`. The model reads every input from that folder and writes its outputs back into it. Weather data files can be kept in separate folders named in file.cio; everything else is in the one folder.

The master file

file.cio is the index of the project. Each line names a group of inputs and the files to read for it, in a fixed order. `null` means that the file is not used. Read this file first when you open a project. The full description is on file.cio.

file.cio of the `Ames_sub1` reference dataset at 62.0.1 (spacing shortened):

```
file.cio: AMES
simulation      time.sim  print.prt  null  object.cnt  null
basin           codes.bsn  parameters.bsn  carbon.bsn
climate         weather-sta.cli  weather-wgn.cli  null  pcp.cli  tmp.cli  null  null  null  null
connect         hru.con  null  null  null  null  null  null  null  null  null  null  null  null
channel         null  null  null  null  null  null  null  null
reservoir       null  null  null  null  null  null  null  null
routing_unit    null  null  null  null
hru             hru-data.hru  null
exco            null  null  null  null  null  null
recall          null
dr              null  null  null  null  null  null
aquifer         nbull  null
herd            null  null  null
water_rights    null  null  null
link            null  null
hydrology       hydrology.hyd  topography.hyd  null
structural      tiledrain.str  septic.str  filterstrip.str  grassedww.str  bmpuser.str
hru_parm_db     plants.plt  fertilizer.frt  tillage.til  pesticide.pes  null  null  null  urban.urb  septi
ops             harv.ops  graze.ops  irr.ops  chem_app.ops  fire.ops  sweep.ops
lum             landuse.lum  management.sch  cntable.lum  cons_practice.lum  ovn_table.lum
chg             null  null  null  null  null  null  null  null  null
init            plant.ini  soil_plant.ini  null  null  null  null  null  null  null  null  null  null
soils           soils.sol  nutrients.sol  null
decision_table  lum.dtl  null  null  null
regions         ls_unit.ele  null  null  null  null  null  null  null  aqu_catunit.ele  null  null  null
pcp_path        null
tmp_path        null
slr_path        null
hmd_path        null
wnd_path        null
```

The `basin` line names three files from revision 62 on (carbon.bsn was added). A file.cio written for an earlier revision has two names there and is misread by 62.0.1; rewrite it with SWAT+ Editor 4 or add the third name. The same applies to the `water_rights` line, which names the water allocation files. See file.cio for the details.

Input groups

Simulation control

File	Content
time.sim	start and end dates, time step
print.prt	which output files to write, and at which time steps
object.cnt	number of objects of each type (HRUs, channels, aquifers, …)
object.prt	extra output for individual objects (optional)

Basin

File	Content
codes.bsn	method switches (PET method, runoff method, routing, carbon model, gwflow, …)
parameters.bsn	basin-wide parameters
carbon.bsn, carbon_lyr.bsn	carbon model parameters; carbon_lyr.bsn is found from the name of carbon.bsn and is not listed in file.cio

Climate

File	Content
weather-sta.cli	weather stations and the data files each one uses
weather-wgn.cli	weather generator statistics, used to fill missing data or generate whole records
pcp.cli, tmp.cli, slr.cli, hmd.cli, wnd.cli	lists of the data files for each variable
*.pcp, *.tmp, *.slr, *.hmd, *.wnd	daily (or sub-daily) records

Objects and connections

File	Content
hru.con, rout_unit.con, aquifer.con, chandeg.con, reservoir.con, recall.con, …	position of each object and where its outflow goes
hru-data.hru	for each HRU, the names of its topography, hydrology, soil, land use, initial conditions, snow and field records
rout_unit.def, rout_unit.ele, ls_unit.def, ls_unit.ele, aqu_catunit.ele	which HRUs belong to each routing unit, landscape unit and aquifer catchment

Hydrology and topography

File	Content
hydrology.hyd	HRU hydrology parameters (lateral flow, evaporation, percolation, …)
topography.hyd	slope, slope length and related values
field.fld	field length, width and orientation

Soils

File	Content
soils.sol	soil layers and their physical properties
nutrients.sol	initial soil nutrient contents

Land use and management

File	Content
landuse.lum	land uses, each naming its plant community, management schedule, curve number set, conservation practice and structural practices
management.sch	management schedules: dated operations and decision tables
cntable.lum, cons_practice.lum, ovn_table.lum	curve numbers, USLE support practice factors, overland Manning's n
lum.dtl	decision tables for land use and management
harv.ops, graze.ops, irr.ops, chem_app.ops, fire.ops, sweep.ops	parameters of each type of operation

Databases

File	Content
plants.plt	plant growth parameters
fertilizer.frt, manure_db.frt, manure_om.frt	fertilizers and manures
tillage.til, pesticide.pes, urban.urb, septic.sep, snow.sno	tillage, pesticide, urban, septic system and snow parameters

Initial conditions

File	Content
plant.ini	plant communities at the start of the run
soil_plant.ini	which soil nutrient and constituent records each HRU starts with

Channels, aquifers, reservoirs and wetlands

File	Content
channel-lte.cha, hyd-sed-lte.cha, nutrients.cha, initial.cha	channels (the SWAT-DEG channel model, connected through chandeg.con)
aquifer.aqu, initial.aqu	aquifers
reservoir.res, hydrology.res, sediment.res, nutrients.res, initial.res, weir.res	reservoirs
wetland.wet, hydrology.wet	wetlands

Calibration

File	Content
cal_parms.cal, calibration.cal	parameter changes applied at start-up (Hard calibration)
codes.sft and the other *.sft files	Soft calibration

Optional groups (structural practices, water allocation, gwflow, salts and constituents, pesticides and pathogens, export coefficients, recall) are described in the Input Reference.

Outputs

Output files end in _day, _mon, _yr or _aa (average annual) and .txt, or .csv when CSV output is switched on in print.prt. The first part of the name gives the object (basin_, lsunit_, hru_, channel_sd_, aquifer_, reservoir_, …) and the next part the variables:

Code	Variables
`wb`	water balance
`nb`	nutrient balance
`ls`	sediment and nutrient losses
`pw`	plant growth and stress
`aqu`, `sd_cha`, `res`	aquifer, channel, reservoir

The Output Reference describes each file.

Log and check files

File	Content
`simulation.out`	run progress and the completion message
`diagnostics.out`	names that could not be matched, missing files, unexpected values
`checker.out`	summary of the inputs for each HRU, written before the simulation
`files_out.out`	list of the output files written
`area_calc.out`	area summary
`erosion.out`	header lines only in 62.0.1
`success.fin`	written only when the run completes

Check `simulation.out` and `diagnostics.out` after every first run of a new or changed project.

Sources

- File groups and default names: the `input_*` types in `src/input_file_module.f90`; `file.cio` is read by `src/readcio_read.f90`.
- Example: `refdata/Ames_sub1/file.cio`. The `aquifer` line contains `nbull`, a typing error in the shipped file; the dataset has no aquifers, so the name is never used.

The steps below build a SWAT+ project for an area of your choice, run it and look at the output, with what each step produces. They do not go through QSWAT+ screen by screen.

Requirements

Data:

- A digital elevation model (DEM) covering your area, in a projected coordinate system, with no holes.
- A soil map, either SSURGO (US), HWSD (global), or a custom raster paired with a soil property lookup table.
- A land use map, either NLCD (US), ESA WorldCover (global), or a custom raster paired with a SWAT+ land use lookup table.
- A daily climate record (precipitation and temperature at minimum, optionally solar radiation, wind, humidity) from one or more stations covering the simulation period.
- An outlet point (the gauge or stream location you want to drain to).

Software:

- QGIS with the QSWAT+ plugin, SWAT+ Editor and, for calibration, SWAT+ Toolbox. See Install.

Step 1: Delineate the watershed (QSWAT+)

In QGIS, create a new QSWAT+ project and load the DEM. QSWAT+ will:

1. Fill sinks in the DEM.
2. Compute flow direction and accumulation.

3. Build the stream network at a threshold drainage area you choose.
4. Subdivide the watershed into subbasins at stream junctions.
5. Snap your outlet point to the stream network.

Output: a polygon layer of subbasins and a polyline layer of channels (reaches).

Step 2: Create HRUs

Overlay the land use map, soil map, and a slope class layer on the subbasins. Each unique combination of (subbasin, land use, soil, slope class) is a candidate HRU. QSWAT+ writes the HRUs and the connectivity files.

You can keep all candidate HRUs (high spatial detail, slow run) or filter by area threshold (faster, less detail).

Output: HRU definitions and `*.con` connectivity files.

Step 3: Build the input files (SWAT+ Editor)

Open the project in SWAT+ Editor. The editor will:

1. Import climate station data and write the `.cli`, `.pcp`, `.tmp`, and `weather-wgn.cli` files. Stations without data get filled by the weather generator.
2. Populate soil layer parameters from the soil property table.
3. Populate plant, fertiliser, tillage, and pesticide databases.
4. Let you assign management schedules to each land use.
5. Write the final input folder.

Set the simulation period and choose the outputs. Start the simulation one to three years before the period you want to analyse, and set the number of warm-up years in `nyskip` of `print.prt` so they are left out of the outputs.

Output: a folder of plain-text SWAT+ inputs, ready to run.

Step 4: Run

Run the model from SWAT+ Editor, or from a terminal inside the project's `TxtInOut` folder:

```
cd path/to/your/project/Scenarios/Default/TxtInOut
path/to/swatplus-62.0.1-gnu-lin_x86_64-Rel
```

Watch for the `Execution successfully completed` message. Then read `diagnostics.out`: it lists every name that could not be matched (a management schedule, a weather station, a plant) and every missing file. Fix those before looking at results.

Step 5: Look at output

Start with the basin water balance:

`basin_wb_aa.txt`

The annual-average precipitation, evapotranspiration, surface runoff, lateral flow, percolation, and water yield should be in the range you expect for your climate. If precipitation is far from the expected value, the weather stations are probably assigned wrongly. If ET takes up nearly all of the precipitation, check land use and plant parameters.

Then look at the channel output (`basin_sd_cha_aa.txt` or the per-channel `channel_sd_aa.txt`) for streamflow.

If you have measured streamflow at the outlet, compare to `channel_sd_day.txt` for the gauge channel.

Step 6: Calibrate

The first run is rarely a good match to observed data. Move on to Calibration to adjust parameters and improve the fit.

Common first-run problems

- No precipitation anywhere: the climate station coordinates are outside the watershed extent, so QSWAT+ assigned every HRU to the weather generator instead of the station. Check `weather-sta.cli`.
- Crops never grow: the management schedule is not assigned, or the planting date is outside the simulation period.
- All flow is surface runoff: curve numbers are too high. Check `cntable.lum` and the soil hydrologic group.
- Streamflow is zero downstream: a channel object has no upstream connection in `chandeg.con` or `hru.con`. Inspect the connectivity.

  Sources

  - File names and groups: `src/input_file_module.f90`. Completion message: `src/main.f90.in`.